

## Use and connect to multiple AKS clusters

### az aks get-credentials

- To **connect to an AKS cluster via kubectl**, use the command: ***az aks get-credentials -g <rname> -n <aksname>***
- This command **downloads the kubeconfig file** for the AKS cluster and merges it with the local ~/.kube/config file on your machine
- The **kubeconfig file contains information needed** for kubectl **to communicate** with the Kubernetes **API server** running in the AKS cluster
- The az aks get-credentials command:
  - **retrieves authentication** information from the AKS cluster
  - **creates a new kubeconfig file or merges** AKS cluster information into the existing kubeconfig file
  - **configures kubectl to use the new file**

## Use and connect to multiple AKS clusters

### kubeconfig

- The **kubeconfig** file is a configuration file used by **kubectl** to connect to a **Kubernetes cluster**
- The file **contains information about the cluster**, including the cluster's API endpoint, CA certificate, and authentication credentials
- Kubeconfig files **can contain multiple contexts**, each with its own cluster, user, and namespace configuration
- This **allows users to switch between different Kubernetes clusters** or namespaces easily
- The **kubeconfig** file **contains** the following sections:
  - **Clusters**: defines the Kubernetes clusters that you want to connect to
  - **Users**: defines the users who can authenticate to the Kubernetes cluster
  - **Contexts**: defines the context of the connection, which is a combination of a cluster and a user
  - **Current context**: specifies the current context to use by default when interacting with the Kubernetes cluster. Any commands run against the Kubernetes cluster will use the cluster, user, and namespace defined in the current context